

System that can read closed books

Scientists, including one of Indian origin, have developed a new technology that can read pages of a closed book, an advance that may help archaeologists look into antique books without touching these. Researchers, including Ramesh Raskar from Massachusetts Institute of Technology (MIT), USA, tested a prototype of the system on a stack of papers, each with one letter printed on it. The system was able to correctly identify the letters on the top nine sheets.

The researchers have developed algorithms that acquire images from individual sheets in stacks of paper, and interpret the often-distorted or incomplete images as individual letters.

Terahertz-frequency profiles can distinguish between ink and blank paper, in a way that X-rays cannot, and have much better depth resolution than ultrasound. The system exploits the fact that, between the pages of a book, tiny air pockets are trapped about 20 micrometres deep. The difference in refractive index—the degree to which these bend light—between the air and the paper means that the boundary between the two will reflect terahertz radiation back to a detector.

Octopus-like robot can move without batteries

A team of researchers from Harvard University, USA, has built a soft robot that has no electronics. The robot is powered by a chemical reaction controlled by microfluidics. The team, consisting of members with experience in mechanical engineering, microfluidics and 3D printing, calls their creation the octobot.

Octopuses have no internal skeleton, yet they can perform incredible feats of dexterity and strength. This has made them a long-standing inspiration for soft robotics.

Octobot is small and has been created through 3D printing. It is not tethered to any external hardware and is entirely soft. Researchers believe it will be the forerunner of a new generation of completely autonomous, soft machines. The research has proven that key components of a simple, entirely soft robot can easily be manufactured and that rigid components like electronic controls and batteries can be replaced by equivalent soft systems.



A pneumatic network (red) is embedded within the octobot's body and hyperelastic actuator arms (blue) (Image courtesy: www.sciencenewsjournal.com)

Artificial skin that can send pressure sensation to brain cell

Skin is the largest organ on our body, and the most important interface for human to interact with the outside world. Scientists have always been trying to develop a synthetic skin to mimic the function of natural skin, so that it would not only help us restore the touching ability, but also bring lots of opportunities to develop new materials and electronics for wearable electronics, flexible displays and robotics.



Stanford chemical engineering Prof. Zhenan Bao and her team have created a skin-like material that can tell the difference between a soft touch and a firm handshake. The device on the golden fingertip is the skin-like sensor developed by the engineers (Image credit: Bao Lab, Stanford University)

Engineers at Stanford University, USA, have created a plastic skin that can detect how hard it is being pressed and generate an electric signal to deliver this sensory input directly to a living brain cell.

Zhenan Bao, a professor of chemical engineering at Stanford University, has spent a decade trying to develop a material that mimics skin's ability to flex and heal, while also serving as the sensor net that sends touch, temperature and pain signals to the brain. Ultimately, she wants to create a flexible electronic fabric embedded with sensors that could cover a prosthetic limb and replicate some of skin's sensory functions.

Bao's work takes another step towards her goal by replicating one aspect of touch—the sensory mechanism that enables us to distinguish the pressure difference between a limp handshake and a firm grip.

Ford to mass produce self-driving cars

Ford has said that it will build a totally self-driving car by 2021 as it seeks to take the lead in the global race to produce the world's first high-volume driverless vehicle. The car will have no steering wheel or pedals, and will be used in driverless taxi services.

The carmaker also said that it expected to first roll out the driverless vehicles in big cities. The cars would most likely be hybrids rather than full electric vehicles or cars with a conventional internal combustion engine, to ensure range and minimise downtime needed for recharging.